Enhancing Air Defense Decision Reliability: A Bi-level Cooperative Task Assignment Algorithm for Fortress Protection

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***Abstract*—Weapon target assignment (WTA) is a critical military research topic that can assist commanders in realizing battlefield decisions. This paper investigates a new variant of WTA, called cooperative engagement of weapon systems (CEWS), which achieves defense task assignment through the cooperation of multiple weapon systems, thus significantly improving defense reliability and resource utilization. Considering the decision sequence of optimization objectives, a bi-level optimization model based on CEWS is developed to formulate the problem. Meanwhile, a bi-level task assignment algorithm is proposed for solving the model. Firstly, an improved genetic algorithm is introduced to solve the upper-level model. Secondly, a population initialization method based on combat readiness rules and Tent chaotic mapping, as well as a multi-level enhanced adaptive mutation operator, are designed to facilitate the generation and development of lower-level model solutions. Then, a strategy selection mechanism based on Stackelberg game is designed to assist commanders in making final decisions. Finally, the proposed method is systematically compared with existing heuristic algorithms through multi-scale case studies. The experimental results show that the proposed algorithm can obtain better solutions in most testing scenarios.**

***Index Terms*—Weapon target assignment, denfense reliability, bi-level optimization algorithm, cooperative engagement of weapon systems, Stackelberg game**

I. Introduction

THE weapon target assignment (WTA) problem is a significant branch in military research [1]. Since Manne [2] introduced the WTA problem into the field of operations research, it has attracted widespread attention from military and academic researchers [3]. With the development of automatic control and information technology, swarms, missiles, and guidance systems have become the primary means of attack for various countries [4], [5]. For fortress defense, the cooperation of multi-weapon systems can not only accomplish challenging tasks that cannot be accomplished by a single system, but also provide decision-makers with flexible and reliable solutions [6]. Therefore, researchers are encouraged to explore more advanced cooperative tasking techniques for multi-weapon systems in order to maximize the defense effectiveness of fortresses.

The WTA problem can be categorized into two parts: modeling and algorithm solving. In terms of modeling, it can be divided into two types according to time window constraints: static pre-assignment model and dynamic model. Existing research on WTA issues indicates that dynamic models can be viewed as multi-stage static models. Therefore, studying static pre-assignment models has practical application significance. Whether it is a static pre-assignment model or dynamic assignment model, it usually focuses on the residual value of incoming weapon targets [7], weapon resource consumption [8], weapon killing capability [9], and system efficiency [10], [11]. Contemporary research prioritizes multi-objective optimization frameworks to address complex system requirements [12]. However, despite substantial progress made in multi-objective models, there are still some challenges in practical applications. This limitation stems primarily from hierarchical dependencies among objectives that require sequential prioritization in practical decision-making processes. In order to address this issue, Zhao et al. [13] proposed a bi-level optimization model for WTA, called ECM-DWTA. The upper-level model aims to maximize the strike effect of the entire combat phase, while the lower-level model is dedicated to the global optimization of weapon assignment schemes. ECM-DWTA effectively balances the decision-making sequence between different optimization objectives while ensuring the effectiveness of firepower strikes. Similarly, considering the uncertainty and complexity of the combat environment, Zhu et al. [14] proposed a bi-level anti-missile weapon target assignment model. Compared with ECM-DWTA, this model pays more attention to the cooperation efficiency between weapon systems (WSs), further enhancing the practicality of the model. In addition, Shamami et al. [15] designed a bi-level WTA optimization model, aiming to solve the problem of battlefield information asymmetry and ensure the effectiveness of defense decisions. Although these studies consider the issue of decision order between objectives, they overlook the strong coupling relationship between the upper-level and lower-level models, which may lead to unreliable model solutions and cause difficulty in meeting practical needs. Moreover, existing models fail to focus on maximizing the cooperative engagement capability of weapon systems (CECWS), making it difficult to obtain the global optimal solution for the cooperative engagement of weapon systems (CEWS) problem.

In terms of algorithms, WTA is an NP-complete problem [16], and traditional exact algorithms face significant difficulties in solving it. In contrast, some intelligent methods, such

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as heuristic algorithms and deep reinforcement learning, have significant advantages in solving such problems. Recently, researchers have attempted to use heuristic algorithms such as algorithm of genetic (GA) [17], improved particle swarm optimization [18], improved firefly [19], Gaussian mutation beetle swarm optimization [20], artificial bee colony (ABC)[21], and multi-objective artificial bee colony (MOABC) [22] to solve the WTA problem. It is worth noting that heuristic algorithms are currently the main means of solving real-world engineering optimization problems [23], [24], [25], [26]. Among them, GA, ABC, and MOABC have favorable scalability and flexibility due to their simple structure, strong robustness, and ease of integration with other optimization algorithms, which is crucial for solving WTA problems. Especially, the evolutionary operators of GA have a natural advantage in dealing with binary-coded problems, while swarm intelligence algorithms such as MOABC are more suitable for dealing with real-coded problems. Therefore, the combination of evolutionary algorithms and swarm intelligence algorithms can flexibly address WTA problems with complex decision relationships. In addition, Guo et al. [27] proposed a Q-learning based WTA solving algorithm. Compared with heuristic algorithms, this algorithm has significant advantages in computational speed. However, the Q-learning method requires maintaining a Q-value table, and as the problem size increases, the dimensions of the Q-value table will grow exponentially, leading to a dimension disaster. In contrast, the WTA solving algorithm based on deep Q-learning proposed by Luo et al. [28] utilized deep neural networks to approximate Q values, thereby avoiding the curse of dimensionality and significantly improving the algorithm's generalization ability. However, the widespread application of data-driven methods is constrained by challenges such as obtaining valid samples, overfitting, and poor interpretability. For bi-level WTA, a series of algorithms have been proposed, including the hybrid multi-objective bi-level interactive fuzzy programming algorithm [13], the improved bi-level recursive biogeography-based optimization algorithm [14], and the heuristic combination algorithm [15]. These works provide a theoretical foundation for addressing this class of optimization problems. However, when solving the CEWS problem, the cooperative relationships between WSs are unconstrained. This can produce solutions that violate cooperative constraints, thereby rendering the assignment scheme ineffective.

In summary, the WTA problem has made some achievements in modeling and solving algorithms. However, there are still several issues. Specifically, the CEWS typically exhibits a clear hierarchy in actual command decision-making. Senior commanders focus on the improvement of overall defense effectiveness, while grassroots units aim at targets residual threats and resource consumption. Although single-level optimization models can coordinate multiple objectives at the same level, they overlook the decision-making sequence and constraint relationships between upper-level and lower-level objectives. At the same time, these constraint relationships and decision sequences are difficult to clearly express in single-level optimization models, while the bi-level model can naturally model this hierarchical decision-making process and clarify the dependency relationships between the upper-level and lower-level objectives. From an algorithmic perspective, existing optimization algorithms experience a sharp increase in search dimensions when solving CEWS problems, leading to the algorithm easily getting stuck in local optima. Meanwhile, existing optimization algorithms are difficult to handle the interdependence between bi-level tasks, and the solving process can easily damage the cooperative engagement structure, resulting in ineffective cooperative task assignment. In addition, the interpretability of the decision-making process is poor, making it difficult for commanders to determine the optimal solution in the solution set.

To address the aforementioned issues, this paper designs a bi-level optimization model considering CEWS, and proposes a bi-level task assignment algorithm to solve the CEWS model. The main innovations and contributions of this paper can be summarized as follows.

1) A method for calculating the information processing efficiency among WSs is designed. On this basis, a strongly coupled WTA bi-level optimization model considering CEWS and optimization decision sequence is proposed.

2) A bi-level task assignment algorithm (BLTAA) is proposed based on algorithm nesting to decouple the models, enhancing cooperative reliability among multiple weapon systems.

3) To improve BLTAA's search performance, a population initialization method combining combat readiness rules with Tent chaotic mapping (PI-CRRTCM) is developed, along with a multi-level enhanced adaptive mutation operator (MLEAMO).

4) A Stackelberg game-based strategy selection mechanism (SGSSM) is established to support commanders in final decision-making.

The remaining of this paper is organized as follows: Section II introduces the preliminary content. In Section III, a BLTAA is proposed. Section IV presents numerical case verification and analysis. Section V summarizes this paper.

II. Problem Formulation

A. Related work

The WTA issue is one of the most critical components in the command and control system. Li et al. [1] comprehensively analyzed the research results related to WTA in recent years. On this basis, we reviewed representative works from the perspectives of model types, optimization objectives, constraints, and solving algorithms, as shown in Table I. WTA models are divided into two types: single-level models and bi-level models. Among them, single-level models can be further divided into two categories: static WTA models and dynamic WTA models. Although the WTA model is expressed differently, its essence is maximizing our returns. In terms of solving algorithms, heuristic algorithms tend to be applied in static scenarios. The data-driven approach is more suitable for combat scenarios with high real-time requirements.

B. Motivation

When processing CEWS with the existing WTA optimization models, decisions are made only at a single-level, which

TABLE I
VARIOUS WTA MODELS WITH THEIR OPTIMIZATION OBJECTIVES, CONSTRAINTS, AND METHODS

Model type	Optimization objective	Constraint type	Method
Static WTA model [8]	Minimize resource consumption	1. Benchmark constraint 2. Destruction constraint	Adaptive large-scale neighborhood search combined with simulated annealing algorithm
Static WTA model [28]	1. Minimize resource consumption 2. Maximize strike returns	1. Benchmark constraint	Data-driven policy optimization with deep reinforcement learning
Static WTA model [29]	1. Minimize resource consumption 2. Maximize strike returns	1. Benchmark constraint 2. Destruction constraint	Multi-strategy threshold public goods game algorithm
Static WTA model [30]	1. Minimize threat value of the target 2. Minimize time cost	1. Benchmark constraint 2. System constraints	Deep Q-learning network with improved multi-objective artificial bee colony algorithm
Dynamic WTA model [21]	Minimize threat value of the target	1. Benchmark constraint 2. Destruction constraint 3. Time constraint	Improved ABC
Dynamic WTA model [31]	Maximize strike returns	1. Benchmark constraint 2. Feasibility constraints 3. Time constraint	Attention enhanced multi-agent distributional reinforcement learning with dynamic reward algorithm
Dynamic WTA model [32]	1. Maximize strike returns 2. Minimize resource consumption	1. Benchmark constraint 2. Range constraint 3. Destruction constraint	Value iteration algorithm
Dynamic WTA model [33]	Minimize threat value of the target	1. Benchmark constraint	Elite-centered artificial bee colony algorithm with extended solution boundary and adaptive colony scaling
Bi-level WTA model [13]	Upper-level model: Maximize the overall interference effect of the target Lower-level model: 1. Maximize strike returns; 2. Minimize resource consumption	1. Benchmark constraint 2. Time constraint 3. Destruction constraint	Hybrid multi-objective bi-level interactive fuzzy programming algorithm
Bi-level WTA model [15]	Upper-level model: Minimize the degree of target destruction Lower-level model: Maximize the level of regional protection	1. Benchmark constraint 2. Destruction constraint 3. Regional weight constraint	Simulated annealing- Genetic algorithm- Grey wolf optimizer
Bi-level WTA model [14]	Upper-level model: 1. Maximize strike returns 2. Minimize resource consumption Lower-level model: Maximizing cooperative effect	1. Benchmark constraint 2. Time constraint	Improved bi-level recursive BBO algorithm

Note: The benchmark constraint refers to the fundamental conditions or limitations that must be satisfied, as detailed in Ref [2].

makes it difficult to balance global and local demands, resulting in fragmented decision-making logic and unreasonable resource assignment. The bi-level model adopts a layered design. The upper-level is responsible for optimizing the overall combat effectiveness, while the lower-level focuses on resource assignment for execution units. This design can accurately depict the asymmetric game relationship between the global and local, effectively solves the problem of fragmentation in traditional decision-making logic and provides a better solution for CEWS. However, the existing bi-level WTA models have not systematically modeled the CEWS process, thus failing to fully utilize the potential of system defense. Given this consideration, this paper develops a bi-level WTA optimization model based on CEWS to consider the decision priorities of the optimization objectives, as shown in Fig. 1. Specifically, in the scenario of fortress air defense operations, m WSs are engaged in combat with n incoming targets. The upper-level model is constructed as a single-objective optimization framework that maximizes the CECWS through integrating three key operational components: the cooperative killing value of WS on target j ($K_{\mathrm{co},j}$), the cooperative information processing effectiveness ($M_{\mathrm{co},j}$), and the cooperative sensing effectiveness (O_{co}). The upper-level optimization results are used as inputs to drive the lower-level bi-objective optimization model to solve the remaining threat value of the incoming target and the weapon resources consumption. The parameter representations and meanings involved in CEWS are shown in Table II.

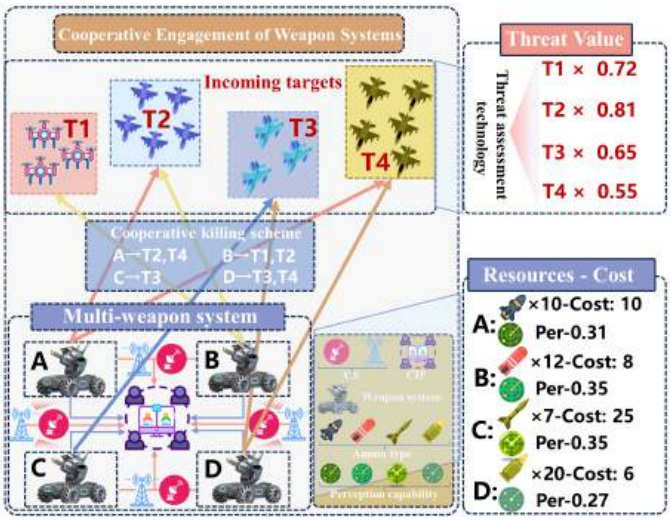


Fig. 1. Schematic diagram of CEWS. Where CS stands for sensing system and CIP indicates cooperative information processing.

C. Upper-level model

1) Modeling of K_{co}: Since there are differences in the strike effectiveness of WS in cooperative engagement, we

TABLE II
PARAMETER REPRESENTATION AND DESCRIPTION OF CEWS MODEL

Parameters	Description
$\mathbf{X}=[x_{ij}]_{m\times n}$	The binary decision variable matrix of the upper-level model; $x_{ij}=1$ indicates that WS i is selected to strike target j; $x_{ij}=0$ indicates that WS i is not selected to strike target j.
$\mathbf{Y}=[y_{ij}]_{m\times n}$	The decision variable matrix of the lower-level model; y_{ij} denotes the quantity of ammunition assigned from WS i to target j.
m	The number of WSs
n	The number of incoming targets
$K_{\text{co},j}$	The cooperative killing value of WSs on target j
$M_{\text{co},j}$	The cooperative information processing effectiveness
O_{co}	The cooperative sensing effectiveness
d	The interval mapping coefficient; $d=6$.
o_{s}	The offset parameter; $o_{\text{s}}=0.5$.
c_{vp}	The central value of cooperation perception; $c_{\text{vp}}=7$.
b_i	The cooperation degree of WS i
$b_{i\text{max}}$	The maximum cooperation degree of WS i in the current environment
$b_{i\text{min}}$	The minimum cooperation degree of WS i in the current environment
s	The baseline of collaborative effectiveness; $s=2/3$.
a_{ij}	The killing effectiveness of weapon i on target j
V_j	The weight of the incoming target j
g	The number of WSs for cooperative processing
M_i	The information processing capability of WS i
ϵ	The gain coefficient
N_j	The number of WSs assigned to target j
R_i	The maximum number of targets that can be assigned to WS i for strike
O_j	The minimum number of cooperative WSs required to successfully strike the target j
E_{sum}	The total value of target threat
β_j	The threat coefficient of incoming target j
P_{ij}	The single-shot kill probability of WS i against target j
C_i	The ammunition cost of WS i
U_i	The number of ammunition possessed by WS i
D_j	The minimum destruction probability for effectively striking incoming target j
t_i	The response time of WS i
$T_{\text{max}}^{\text{sum}}$	The time threshold beyond which the defense system will suffer losses
T_j	The response threshold for a single target

define a cooperation augmentation factor (a_i^{co}) to quantify the improvement in the WS cooperative engagement capabilities. Specifically, parameter a_i^{co} is calculated based on the sigmoid function. The Sigmoid function used in this paper aims to describe the nonlinear saturation relationship between the cooperation degree and combat effectiveness, reflecting the phenomenon of diminishing marginal benefits in the process of enhancing cooperation. When the degree of cooperation increases, the value of a_i^{co} becomes larger, indicating that the cooperating WS can exert stronger combat capabilities. The mathematical expression of a_i^{co} is defined as follows:

$$a_i^{\text{co}}=\frac{1}{1+o_{\text{s}}e^{-\left(\frac{d}{b_{i\text{max}}-b_{i\text{min}}}\left(b_i-\frac{b_{i\text{max}}-b_{i\text{min}}}{2}\right)\right)}}-s \tag{1}$$

CEWS aims to enhance the ability to strike targets, considering the general situation where the more WS assigned to the same target, the stronger its killing ability against that target. Therefore, the expression of $K_{\text{co},j}$ for target j is as follows:

$$K_{\text{co},j}=1-\prod_{i=1}^{m}\left(1-\left(a_{ij}\left(1+a_i^{\text{co}}\right)\right)^{x_{ij}}\right) \tag{2}$$

In particular, if a target is only assigned to one WS, there is no need to calculate the cooperation augmentation factor, and a_i^{co} can be assigned as 0.

Based on Eq. (2), K_{co} for all targets can be expressed as

$$K_{\text{co}}=\sum_{j=1}^{n}V_jK_{\text{co},j} \tag{3}$$

2) Modeling of M_{co}: To simplify the information processing model of WS, this paper considers the information transmission capability (ITC), information processing capability (IPC), and network management capability (NMC) [34]. In fact, when ITC, IPC, and NMC are constant, redundant and overlapping incoming information, as well as the increased number of tasks assigned to WSs, will reduce their information processing capabilities. However, in operational scenarios where multiple WSs are simultaneously processing task information, $M_{\text{co},j}$ enhances the information handling efficiency. Considering the above two situations, the $M_{\text{co},j}$ of WS is designed as

$$M_{\text{co},j}=\left(\prod_{i=1}^{m}M_i^{x_{ij}}\right)^{\frac{1}{g}}\epsilon \tag{4}$$

Compared to a single weapon system, the efficiency of cooperative information processing among multiple weapon systems is higher. Therefore, we define that when $g=1$, $\epsilon=1$; when $g\geq 2$, $\epsilon=1.2$. See Appendix A and C for ITC, IPC, NMC, and M_i.

Considering all targets, the total value of $M_{\text{co},j}$ can be expressed as follows:

$$M_{\text{co}}=\sum_{j=1}^{n}M_{\text{co},j} \tag{5}$$

3) Modeling of O_{co}: Radar is the primary perception equipment of WSs. Whether it is a single WS or CEWS, the radar

with better perception of target j should prioritize detecting the target. This means that in CEWS, the perception capability is mainly determined by the WS with strong perception capability. This is also different from the $M_{\text{co},j}$. Therefore, the O_{co} of WS is designed as

$$O_{\text{co}} = \sum_{j=1}^{n}\left(1-\left(\left(1+e^{\left(-\left(\frac{N_j-c_{\text{vp}}}{2}\right)^2\right)}\right)\cdot\max\{x_{ij}R_{\text{ar}}^{ij}\}\right)\right) \tag{6}$$

See Appendix B and C for R_{ar}^{ij}.

4) Objective function and constraints: The objective of the upper-level model is to maximize CECWS. Based on the above content, the upper-level objective function is expressed as:

$$\max f_1^{\text{upper}}(\mathbf{X}) = K_{\text{co}}(\mathbf{X})M_{\text{co}}(\mathbf{X})O_{\text{co}}(\mathbf{X}) \tag{7}$$

The general constraints of the upper-level model are summarized as

$$\sum_{j=1}^{n} x_{ij} \le R_i, \forall i = 1,2,...,m \tag{8}$$

$$\sum_{i=1}^{m} x_{ij} \ge O_j, \forall j = 1,2,...,n \tag{9}$$

$$\sum_{j=1}^{n} x_{ij} \ge 1, \forall i = 1,2,...,m \tag{10}$$

D. Lower-level model

After determining the cooperative strike scheme in the upper-level model, it is necessary to optimize the assignment of weapon and equipment quantities to minimize the remaining threat values of incoming targets and minimize resource consumption. The objective function of minimizing the remaining threat values of the incoming targets is represented as follows:

$$\min f_1^{\text{lower}}(\mathbf{Y};\mathbf{X}) = E_{\text{sum}} - \sum_{j=1}^{n}\beta_j\left(1-\prod_{i=1}^{m}(1-P_{ij})^{y_{ij}x_{ij}}\right) \tag{11}$$

The resource consumption objective function, composed of weapon acquisition costs and assignment quantities, is formulated as follows:

$$\min f_2^{\text{lower}}(\mathbf{Y};\mathbf{X}) = \sum_{j=1}^{n}\sum_{i=1}^{m}(C_i y_{ij})^{x_{ij}} \tag{12}$$

The general constraints for the lower-level model are as follows:.

$$\sum_{j=1}^{n} y_{ij} \le U_i, \forall i = 1,2,...,m \tag{13}$$

$$\left(1-\prod_{i=1}^{m}(1-P_{ij})^{y_{ij}x_{ij}}\right) > D_j, \forall j = 1,2,...,n \tag{14}$$

$$\sum_{i=1}^{m}\sum_{j=1}^{n}(t_i y_{ij})^{x_{ij}} \le T_{\max}^{\text{sum}} \tag{15}$$

$$\sum_{i=1}^{m}(t_i y_{ij})^{x_{ij}} \le T_j, \forall j = 1,2,...,n \tag{16}$$

E. Bi-level optimization model representation

Based on the above content, the bi-level optimization model proposed in this paper can be expressed as follows:

$$\begin{aligned} &U_{\text{M}}\left\{\text{Eq.}(7)\middle|\max_{\mathbf{X}} f^{\text{upper}}(\mathbf{X}, L_{\text{M}}(\mathbf{X}))\right\} \\ &L_{\text{M}}(\mathbf{X}) = \arg\min_{\mathbf{Y}}\{\text{Eqs.}(11),(12)\mid \mathbf{X}\text{ fixed}\} \end{aligned} \tag{17}$$

where U_{M} is the upper-level model and $L_{\text{M}}(\mathbf{X})$ is the lower-level model.

III. Methodology

The decision variables of the upper-level model in the proposed CEWS bi-level model are binary, while those of the lower-level model are integers. However, a single heuristic algorithm with discrete integer encoding struggles to handle both variable types simultaneously. To address this, we propose a BLTAA, where the upper-level algorithm employs binary matrix encoding to clarify cooperative relationships, and the lower-level algorithm achieves precise assignment via discrete integer matrix encoding. Through nested algorithm combination, BLTAA enables the separation and solving of bi-level models, ensuring that the lower-level solution always depends on the upper-level solution. Additionally, to select the final decision scheme from the Pareto solution set, we develop an SGSSM method to assist commanders in making optimal decisions. Together, these methods enhance the reliability of fortress defense systems.

A. Upper-level algorithm

Existing algorithms mostly explore the solution space by changing the numerical magnitude of the decision variables, which has limitations when dealing with binary matrix encoding forms. GAs show advantages in this regard as they rely on crossover and mutation between chromosomes rather than change in values. This paper designs an improved genetic algorithm (IGA) to solve the upper-level model. Although GA demonstrate significant advantages in solving binary coding problems, their reliance on fixed parameters during population evolution leads to deficiencies in maintaining diversity, potentially resulting in local optima. Based on previous research results [35] (Appendix D), we introduce an adaptive operator based on truncated mean (AOTM) and design an improved genetic algorithm (IGA) to enhance the algorithm's ability to escape local optima.

B. Lower-level algorithm

Lower-level model optimization is more complex than upper-level model optimization, and it needs to be iteratively solved based on the results of the upper-level model. Traditional evolutionary algorithms have limited effectiveness in this situation. Especially their crossover operators can disrupt the results of the upper-level model. In contrast, the individual position update mechanism of swarm intelligence algorithms, such as MOABC [36] and particle swarm optimization [37], generally does not interfere with the optimization results of upper-level models. Of these, MOABC has significant

potential for application in WTA problems due to its strong robustness and good balance between exploration and exploitation [38]. Based on the MOABC framework, two performance enhancement strategies (PI-CRRTCM and MLEAMO) are designed to solve the lower-level model, aiming to enhance the convergence of the algorithm in solving high-dimensional discrete nonlinear integer programming models. The principles of PI-CRRTCM and MLEAMO are shown in Fig. 2. We tend to name this lower-level solving algorithm as improved adaptive MOABC (IAMOABC).

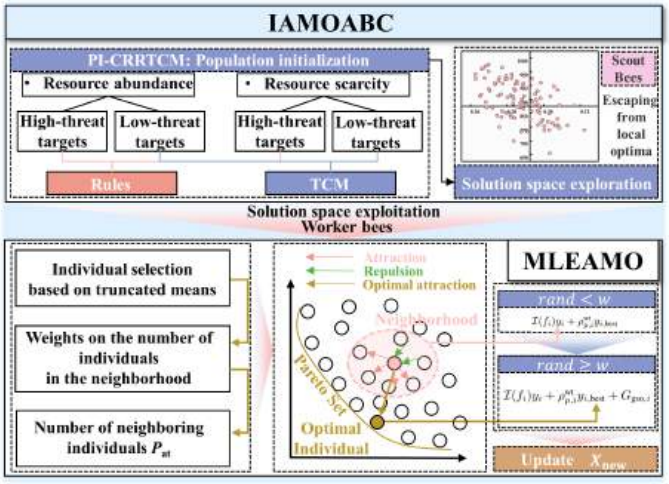


Fig. 2. The implementation principles of PI-CRRTCM and MLEAMO. TCM is a Tent chaotic mapping

1) PI-CRRTCM: The diversity of the initial population ensures that the optimization algorithm can fully explore the vast search space. A reasonable initialization method not only increases the coverage of the solution space, but also improves the global search capability. In the MOABC algorithm, the exploration strategy maintains the diversity of the population by reinitializing the inferior individuals. However, when facing discrete high-dimensional problems, the original initialization method often results in a small and densely distributed solution space, which limits the effectiveness of the algorithm. Moreover, the specific assignment of weapon quantities in real combat is influenced by factors such as ammunition reserves and incoming targets. To address these challenges, this paper designs a PI-CRRTCM method to ensure the population diversity. Operational scenarios are classified into two distinct combat readiness rules: resource abundance (RA) and resource scarcity (RS), through systematic analysis of munition stockpile status, with subsequent mission-specific initialization methods being implemented accordingly.

For high threat targets, WS with higher destruction probabilities are prioritized. If multiple WS have the same destruction probability, assignment is determined based on resource conditions: Under the RA rule, WS with faster response times are prioritized, while under the RS rule, WS with greater ammunition reserves are prioritized. Low threat targets are randomly assigned based on Tent chaotic mapping, defined as follows:

$$K_{i+1} = \begin{cases} \frac{K_i}{a^{\text{pi}}}, & K_i < a^{\text{pi}} \\ \frac{1-K_i}{1-a^{\text{pi}}}, & K_i \geq a^{\text{pi}} \end{cases} \tag{18}$$

where K_i is a random number within the range $[0, 1]$; $a^{\text{pi}} \in (0, 1)$.

The above initialization method using combat readiness rules can adapt to resource constraints, making the initialization solution more suitable for practical application scenarios. Tent chaotic mapping introduces nonlinear dynamic changes, which can ensure diversity during the population initialization process. The proposed initialization method is shown in Algorithm 1.

Algorithm 1: PI-CRRTCM

Input: Upper-model result structures S_t, number of targets n, threat value matrix $\mathbf{T}_v$, weapon system ammunition resource vector $\mathbf{w}_{\text{R}}$, resource threshold R_m, threat threshold H_t

Output: Assignment result structure G

$G = S_t$ // Assignment of the number of munitions on the basis of S_t
for $i = 1 \rightarrow n$ **do**
  if sum$(W_R) > R_m$ **then**
    M = 1 // Resource abundance
  else
    M = 0 // Resource scarcity
  if $T_v(i) > H_t$ **then**
    if $M == 1$ **then**
      WS selection follows RA;
    else
      WS selection follows RS;
    The selected WS prioritizes the assignment of weapons and maps the results to G;
  else
    Map random assignment result to G;
  Update the assignment result to G
$L_T =$ find$(T_v < H_t)$ // Find low threat targets
for $k = 1 \rightarrow$ length(L_T) **do**
  Assignment weapons based on Eq. (18);
  Map the assignment results to G;
  Update the assignment result to G;

Output: Assignment result structure G

2) MLEAMO: CEWS models are usually high-dimensional discrete integer nonlinear and the individual position update strategy of MOABC is prone to getting stuck in local optima, resulting in poor solution convergence. In contrast, the attraction-repulsion optimization algorithm (AROA) proposed by Cymerys et al. [39] drives individual position update by considering the influence between individual neighborhoods, thus possessing strong solution space exploitation capabilities. However, in the original AROA individual position update strategy, the number of individuals in a neighborhood is

updated based on the iterative process. This calculation method can improve the computational efficiency, but it does not consider the relationship between individuals and populations. When facing complex problems, excessively reducing the influence of the neighborhood will limit its exploitation ability. To address this issue, an AROA-based MLEAMO is proposed. MLEAMO drives individual position update by characterizing individual-population relationships to compute neighborhood ranges, determining optimal individual attraction effects and neighborhood attraction effect weights, and combining them with global population individual positions. This updating strategy enhances the exploitation capability of algorithms that avoid prematurely falling into local optima, allowing them to better optimize solutions.

To reasonably characterize individual-population relationships, population individuals are screened using truncated means to eliminate undesirable individuals, denoted as

$$M_{\text{eli}} = M_{\max}\left(\frac{t}{T_{\max}^{\text{L}}}\right)^2\left(1+\alpha\left(\frac{1}{2}-N_{\text{low}}(t)\right)\right) \tag{19}$$

where M_{eli} is the number of individuals eliminated; $M_{\max}$ is the maximum number of single elimination; t is the number of current iterations; $T_{\max}^{\text{L}}$ denotes the maximum number of iterations for the IAMOABC; α is the influence coefficient of population aggregation; $N_{\text{low}}(t)$ is the degree of population dominance, defined as

$$N_{\text{low}}(t) = \frac{n(Pareto,t)}{N_p} \tag{20}$$

where $n(Pareto,t)$ is the number of non-dominated solutions in the population individuals; N_p is the number of populations.

The average fitness of the screened population can be expressed as

$$f_{\text{m}}^*(t) = \frac{\sum_{i=1}^{N_p-M} f_i(t)}{N_p - M} \tag{21}$$

where $f_i(t)$ is the individual fitness.

The individual-population relationship can be characterized by measuring the difference between individual fitness and population fitness and the expression for the weight of the number of individuals in a neighborhood is further proposed and calculated as

$$W_i(t) = \begin{cases} W_{\max}, & f_i(t) < f_{\text{m}}^*(t) \\ W_{\max} - \dfrac{W_{\max}-W_{\min}}{1+e^{(-B_1\Delta(t)-B_2)}}, & f_i(t) \geq f_{\text{m}}^*(t) \end{cases} \tag{22}$$

where $W_{\max}$ is the maximum weight value; W_n is the minimum weight value; B_1 is the slope; B_2 is a constant parameter; $\Delta(t) = (f_i(t) - f_{\text{m}}^*(t))/(f_{\text{m}}^*(t) - f_{\text{n}}^*(t))$; $f_n^*(t)$ is the minimum fitness value.

Based on $W_i(t)$, the proposed mechanism for determining the number of neighboring populations governed by the individual-population relationship is expressed as follows:

$$P_{\text{at}} = A_P + \left\lceil C_p \frac{T_{\max}^{\text{L}} - t}{T_{\max}^{\text{L}}}(1+W_i(t)) \right\rceil \tag{23}$$

where A_P denotes the threshold for the number of individuals in the neighborhood; C_p is a constant parameter. The mechanism preserves the characteristics of the original algorithm and also assesses the relationship between individuals and the population, thus more accurately calculating the number of individuals within the neighborhood.

After determining the number of individuals in the neighborhood, we determine the neighborhood range based on the distance between individuals. According to the characteristics of the coding in this paper, the distance $d_{i,P}$ between individual $\boldsymbol{\mathcal{X}}_i$ and neighborhood individual $\boldsymbol{\mathcal{X}}_p$ is calculated using the Frobenius norm.

$$d_{i,P} = \| \boldsymbol{\mathcal{X}}_i - \boldsymbol{\mathcal{X}}_p \|_F = \sqrt{\sum_{a,b}(x_{ab}^i - x_{ab}^P)^2} \tag{24}$$

The specific location of the neighboring individuals can be confirmed based on $d_{i,P}$.

For the individual position to be updated, the attraction-repulsion mechanism of the neighborhood is calculated as follows:

$$\boldsymbol{\mathcal{Y}}_i = \frac{1}{P_{\text{at}}}\sum_{P=1}^{P_{\text{at}}} c(\boldsymbol{\mathcal{X}}_p - \boldsymbol{\mathcal{X}}_i) I(d) s(f_P, f_i) \tag{25}$$

$$I(d) = 1 - \frac{d_{i,P}}{d_{i,\max}} \tag{26}$$

$$s(f_P, f_i) = \begin{cases} -1, & f_P < f_i \\ 0, & f_P = f_i \\ 1, & f_P > f_i \end{cases} \tag{27}$$

where $\boldsymbol{\mathcal{Y}}_i$ is the neighborhood effect of individual i to be updated; c is the step size; $I(d)$ is a function that evaluates the influence of p-th solution; $s(f_P, f_i)$ is the update direction function; f_P is the fitness of the pth individual in the neighborhood.

In addition, the individual position to be updated should also consider the attraction effect of the optimal individual to it. Therefore, we design an optimal individual attraction strategy, whose strategy function is as follows:

$$\boldsymbol{\mathcal{Y}}_{i,\text{best}} = \begin{cases} c(\boldsymbol{\mathcal{X}}_{\text{best}} - \boldsymbol{\mathcal{X}}_i), & \text{if } rand \geq b_{\text{mel}} \\ cLevy(dim)(\boldsymbol{\mathcal{X}}_{\text{best}} - \boldsymbol{\mathcal{X}}_i), & \text{if } rand < b_{\text{mel}} \end{cases} \tag{28}$$

where $\boldsymbol{\mathcal{Y}}_{i,\text{best}}$ represents the effect of the optimal individual on updating individual i; $Levy(dim)$ is the Levy flight random number; $rand$ denotes a uniformly distributed random number in [0, 1]; b_{mel} is a constant parameter. Over-attraction is prevented by introducing a random factor.

To balance the relationship between the best individual attraction effect and the neighbor effect, an effect-balancing adaptive weight $\rho_{p,i}^{\text{wt}}$ is further proposed. $\rho_{p,i}^{\text{wt}}$ determines its weight by measuring the relationship between the individual to be updated and the neighboring individuals. The specific expression is as follows:

$$\rho_{p,i}^{\text{wt}} = 1 - \left(\mu\left(\frac{1}{P_{\text{at}}}\sum_{P=1}^{P=P_{\text{at}}}\varepsilon(f_P - f_i)\right) + \rho\right) \tag{29}$$

In the equation, $\varepsilon(f_P - f_i)$ is the indicator function. If $f_P \geq f_i$, then $\varepsilon(f_P - f_i) = 1$. If $f_P < f_i$, then $\varepsilon(f_P - f_i) = 0$. μ and ρ are the equilibrium influence coefficients. By measuring

the fitness relationship between individuals and neighboring individuals and adaptively adjusting the weights of $y_{i,\text{best}}$ and y_i, excessive indulgence in a certain attraction mechanism can be prevented.

To further enhance the exploitation capability of the algorithm, based on the population individual position mean, a global search operator (GSO) is designed, which is stated as

$$\mathbf{G}_{\text{gso},i} = \left(X_i - \frac{\sum_{i=1}^{N_p - M} X_i}{N_p} \right) R \tag{30}$$

where R denotes a uniformly distributed random number in [0, 1]. $\mathbf{G}_{\text{gso},i}$ guides the exploitation of solutions by measuring the global differences between individuals and populations.

By setting $\mathcal{I}(f_i) = \left(\mu \left(\frac{1}{P_{\text{at}}} \sum_{P=1}^{P=P_{\text{at}}} \varepsilon(f_P - f_i) \right) + \rho \right)$, synthesizing Eqs. (19)-(30), and introducing GSO into the attraction-repulsion mechanism to enhance the ability of the algorithm to jump out of the local optimum, we obtain the proposed MLEAMO expression as follows:

$$\boldsymbol{\mathcal{X}}_{\text{new}} = \begin{cases} \boldsymbol{\mathcal{X}}_i + \mathcal{I}(f_i)\boldsymbol{\mathcal{Y}}_i + \rho_{p,i}^{\text{wt}} \boldsymbol{\mathcal{Y}}_{i,\text{best}}, & rand < w \\ \boldsymbol{\mathcal{X}}_i + \mathcal{I}(f_i)\boldsymbol{\mathcal{Y}}_i + \rho_{p,i}^{\text{wt}} \boldsymbol{\mathcal{Y}}_{i,\text{best}} + \mathbf{G}_{\text{gso},i}, & rand \geq w \end{cases} \tag{31}$$

The parameter w is a predefined constant.

The MLEAMO dynamically adjusts the influence of neighborhoods on individuals in the population during the search process, ensuring effective exploration in each stage of iterative computation. By introducing random numbers and incorporating global population information into the search process, it effectively improves the algorithm's ability to escape local optima.

C. SGSSM

Contrary to traditional multi-objective optimization approaches that focus solely on finding the Pareto frontier, our model additionally determines the optimal assignment strategy. In CEWS, the upper-level model dominates, while the lower-level model derives its optimal solution from the upper-level's results. Structurally, both models conform to the Stackelberg game framework for military applications [40], [41], [42]. We therefore design a SGSSM, which effectively concatenates model solutions to determine the final cooperative engagement strategy.

Firstly, we define the upper-level optimization algorithm as the leader and the lower-level optimization algorithm as the follower. Secondly, the leader's reward function should consider the follower's payoff. To address this, the leader reward coefficient was designed based on the membership function theory. The leader reward function U^{Fup} of this paper is expressed as follows:

$$U^{\text{Fup}} = \left(1 + \exp\left(-\frac{\sum_{j=1}^{sum} f_{\text{nor},j}^{\text{lower}}}{4\phi^2} \right) \right) Fit_{\text{upper}} \tag{32}$$

where $f_{\text{nor},j}^{\text{lower}}$ is the normalized lower-level model objective function value; sum is the number of objective functions; Fit_{upper} denotes the upper-level model fitness; ϕ is a constant parameter.

Considering the feature of the optimization objective of the lower-level model, the following reward function is simplified and designed as follows:

$$F^{\text{Flo}} = 1 - \sum_{j=1}^{sum} k_j f_{\text{nor},j}^{\text{lower}} \tag{33}$$

where k_j is the weight coefficient.

Finally, according to the above parameter definitions, a Stackelberg game model is established as:

$$\begin{aligned} &\max_{p,a} \sum_{i\in I}\sum_{j\in J}\sum_{k\in K} \tau^k U_{ij}^{\text{Fup}} p_i a_j^k \\ &\text{s.t.} \quad a^k \in \arg\max_{l^k} \left\{ \sum_{i\in I}\sum_{j\in J} F^{\text{Flo}} p_i l_j^k \right\} \\ &\quad\; l_j^k \in \{0,1\} \\ &\quad\; \sum_{j\in J} l_j^k = 1 \end{aligned} \tag{34}$$

where I and J are the leader and follower strategy sets, respectively. In this paper, the top 2 individuals with fitness values are selected as the set of strategies, $I = J = 2$; K is the follower set, $K = 1$; U_{ij}^{Fup} is the reward value when the leader chooses strategy i and the follower chooses strategy j; p and a are the probability distributions of the strategies of the leader and follower, respectively; p_i is the probability of the leader choosing strategy i; a_j^k is the probability of the follower K choosing strategy j; a^k is a mixed strategy of follower K; τ^k is the probability of the leader facing the follower K; l_j^k is the state of the follower K choosing strategy j; l^k is the pure strategy of follower K.

In the bi-level iteration process of BLTAA, based on the calculation results of Eqs. (32) and (33), the strategy combination for each iteration is selected through the Stackelberg game model. Specifically, after bi-level iterations of T, T strategy combinations are obtained. Each combination contains two reward function values (U^{Fup} and F^{Flo}), which are solved by Stackelberg game. Normalize U^{Fup} and F^{Flo} to obtain V_{i1} and V_{i2}, respectively, with $i = 1, \ldots, T$. A decision matrix $\mathbf{D}_{T\times 2}$ is constructed based on V. The optimal strategy combination (OSC) is obtained by calculating $\mathbf{D}_{T\times 2}$ using the TOPSIS method. It should be noted that OSC contains the optimal solutions of the upper-model and the lower-model, making it a unique solution rather than a solution set. The decision matrix $\mathbf{D}_{T\times 2}$ is represented as

$$\mathbf{D}_{T\times 2} = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \\ \vdots & \vdots \\ V_{T1} & V_{T2} \end{pmatrix} \tag{35}$$

The implementation process of SGSSM is shown in Fig. 3.

Compared with the TOPSIS method, the SGSSM method can integrate subjective intentions and objective weights more flexibly due to the existence of the reward function. When dealing with multi-objective problems, the SGSSM method can consider multiple factors comprehensively, generating more overall decision-making solutions.

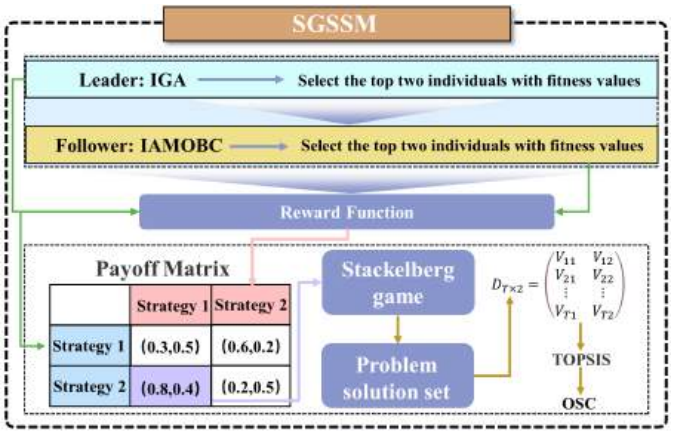


Fig. 3. OSC screening process based on SGSSM

To sum up, this paper proposes a nested combination of IGA and IAMOABC. IGA solves the upper-level model and outputs the cooperative engagement matrix. Based on this, IAMOABC explores and develops solution space by changing the size of non-zero elements in the matrix to achieve resource assignment. During this process, IAMOABC will not alter the cooperative relationship between WS, ensuring the structural correlation between the upper-level and lower-level models. Moreover, to improve the computational efficiency of BLTAA, iterative adaptive truncation strategies are designed for IGA and IAMOABC, respectively. In IGA, if the improvement rate of the most recently f-generation optimal fitness value is lower than threshold U_t or the number of iterations reaches $T^{\mathrm{U}}_{\max}$, the evolution process will be truncated (defined as F1 rule). In IAMOABC, if the continuous h-generation hypervolume change is lower than threshold L_t or the number of iterations reaches $T^{\mathrm{L}}_{\max}$, the computational process will be truncated (defined as H1 rule). By adaptively adjusting the number of iterations, the efficiency of the algorithm can be improved. The calculation implementation process of BLTAA is shown in Fig. 4.

D. Computational complexity analysis

The computational complexity of BLTAA is mainly determined by its bi-level nested structure, which consists of IGA and IAMOABC. The worst computational complexity of IGA is $O\left(T^{\mathrm{U}}_{\max}(N_p mn + N_p \log N_p)\right) = O\left(T^{\mathrm{U}}_{\max} N_p mn\right)$. The main components of IAMOBC computational complexity include fitness calculation, non-dominated sorting, and MLEAMO. The computational complexity of the first two parts is $O(N_p mn)$ and $O(KN_p^2)$, respectively. K is the number of objective functions. The computational complexity of MELAMO can be divided into three parts: individual screening and truncated mean calculation, neighborhood range confirmation, and neighborhood effects. The computational complexity of these three parts is $O(N_p \log N_p)$, $O(N_p^2 mn + N_p^2 \log N_p) = O(N_p^2 mn)$, and $O(N_p^2 mn)$, respectively. Therefore, the worst-case scenario for IAMOABC computational complexity is $O\left(T^{\mathrm{L}}_{\max} N_p^2 mn\right)$. Furthermore, the computational complexity of SGSSM is $O(T)$. In summary, the worst-case scenario for BLTAA computational complexity is $O(T[O(\mathrm{IGA}) + O(\mathrm{IAMOABC})]) + O(T) = O(TT^{\mathrm{L}}_{\max} N_p^2 mn)$.

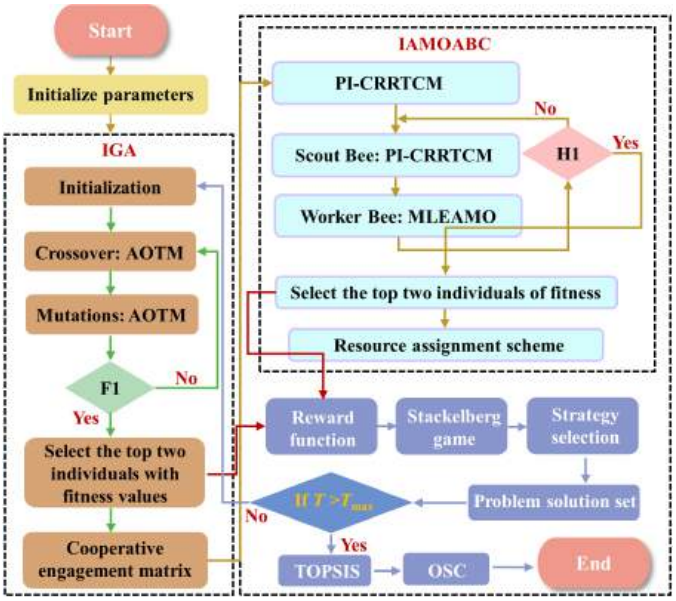


Fig. 4. Flowchart of BLTAA implementation for solving CEWS models, where $T_{\max}$ is the maximum number of bi-level iterations.

IV. Simulation and Results

Due to the sensitivity of the WTA problem, there is currently a lack of publicly available standard datasets for algorithm validation. For this reason, this paper refers to the simulation framework established by the existing research [8], [43] and designs eight different scale defense battle scenarios to comprehensively evaluate the effectiveness and progressiveness of the BLTAA method. The specific scenario parameters are set according to Refs. [8], [19], [20]. In these scenarios, the numbers of WS are 5, 5, 8, 10, 15, 15, 20, and 30, respectively, while the numbers of targets are 5, 10, 8, 12, 15, 20, 25, and 30, respectively. For example, 5×10 represents a scenario with 5 WSs and 10 incoming targets.

The test case generator is used to generate key parameters such as weapon kill probability and target destruction threshold, with the specific value ranges shown in Table III. In terms of algorithm selection, this paper selected classic algorithms closely related to WTA problem and multi-objective optimization algorithms proposed in recent years to ensure comparability in diversity and convergence. The parameter settings of all comparison algorithms are kept consistent with the original literature to ensure fairness and reproducibility of the comparison. The parameter values of BLTAA are as follows: $N_p = 200$, $T^{\mathrm{U}}_{\max} = 200$, $T^{\mathrm{L}}_{\max} = 200$, $T_{\max} = 10$, $\alpha = 0.5$, $A = B = 0.3$, $C_P = 6$, $A_P = 12$, $c = 0.88$, $\mu = 0.34$, $\rho = 0.12$. To ensure the rationality of the BLTAA parameter settings, the criteria for parameter selection are detailed in Appendix E. All experiments are validated through the MATLAB R2021a environment on a 64-bit laptop equipped with a 2.30 GHz Core(TM) i7-12700H CPU and 16 GB of RAM.

TABLE III
PARAMETER VALUE RANGES

Parameters	Value lower limit	Value upper limit
p_{ij}	0.10	0.80
C_i	10	60
t_i	1	8
a_i^{co}	-0.10	0.40
a_{ij}	0.20	0.60
M_i	0.40	0.60
R_{ar}^{ij}	0.40	0.60
E_{sum}	1	2
U_i	6	16
R_i	1	4
O_j	1	4

A. IGA validation

To verify the effectiveness of IGA, a comparison is made with GA, improved adaptive differential evolution (IADE) [44], and improved firefly algorithm (IFA) [19]. DE is similar to GA in solution updating; while IFA relies on population individual value variations. For binary encoding in IFA, values > 0.5 are set to 1 and values < 0.5 are set to 0. The convergence results of the four algorithms are shown in Fig. 5. Meanwhile, each algorithm independently runs 30 times, and the resulting fitness values are subjected to Wilcoxon rank-sum test and Friedman test, as shown in Table IV. In the Wilcoxon rank-sum test, the symbols $+$, $-$, and $\approx$ indicate that the proposed algorithm is significantly superior, significantly inferior, or shows no significant difference compared to the benchmark algorithms, respectively. A smaller Friedman test value represents a higher ranking and better performance of the algorithm.

As shown in Fig. 5, the IGA achieves better solution fitness values across all eight scenarios. The IFA results demonstrate that the individual position update mechanism relying solely on modifications of decision variables is unsuitable for solving the upper-level model. The results in Table IV indicate that IGA has significant advantages in convergence and strong robustness. These findings confirm that the proposed IGA fulfills the design requirements.

B. PI-CRRTCM validation

A reasonable population initialization strategy is crucial for maintaining population diversity and improving algorithm convergence performance. To validate the effectiveness of the proposed initialization method (PI-CRRTCM), the moment of inertia $I_C(t)$ [45] and hypervolume (HV) are employed to evaluate population diversity and the comprehensive performance (convergence and diversity) of the algorithm. The calculation of $I_C(t)$ is as follows

$$I_C(t) = \sqrt{\sum_{i=1}^{N_p}\sum_{d=1}^{D}\left(X_{id}(t) - c_d(t)\right)^2} \tag{36}$$

$$c_d(t) = \frac{1}{D}\sum_{i=1}^{N_p} X_{id}(t) \tag{37}$$

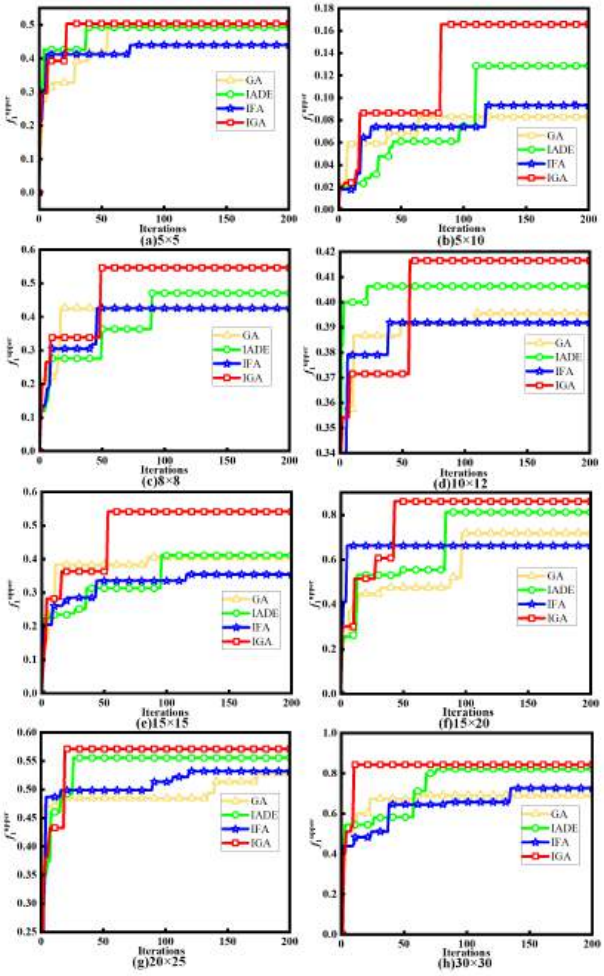


Fig. 5. Comparison result of algorithms at different scenarios

TABLE IV
WILCOXON RANK-SUM TEST AND FRIEDMAN STATISTICAL ANALYSIS RESULTS FOR FITNESS VALUES OF FOUR ALGORITHMS

	GA	IADE	IFA	IGA
5×5	0.4844≈ (0.0288)	0.4811≈ (0.0250)	0.4807- (0.0274)	**0.4944** (**0.0099**)
5×10	0.1119- (0.0423)	0.1139≈ (0.0369)	0.0929- (**0.0231**)	**0.1238** (0.0423)
8×8	0.4758- (0.0936)	0.5058- (**0.0497**)	0.4495- (0.1329)	**0.5214** (0.0922)
10×12	0.4073- (0.0232)	0.4145- (0.0129)	0.3958- (0.0177)	**0.4152** (**0.0114**)
15×15	0.3791- (0.0653)	0.4167- (0.0614)	0.3607- (0.0753)	**0.4321** (**0.0117**)
15×20	0.7220- (0.0959)	0.7622- (0.1259)	0.7069- (0.1033)	**0.8219** (**0.1011**)
20×25	0.5229- (0.0451)	0.5477- (0.0375)	0.5102- (0.0524)	**0.5651** (**0.0337**)
30×30	0.6574- (0.0454)	0.7781- (0.0227)	0.6377- (0.0328)	**0.8015** (**0.0211**)
+/-/≈	0/7/1	0/5/3	0/8/0	
Ranking	1.8889	2.5556	4	1

Note: The upper value in each cell represents the mean, while the value in parentheses represents the standard deviation.

where D is the dimension of decision variables; X_{id} represents the value of the d-th dimension for the i-th population individual at iteration t. The calculation of HV is as follows

$$HV(A,P) = \bigcup_{a_{\text{hv}} \in A}^{A} v(a_{\text{hv}}, P) \tag{38}$$

where A is the Pareto solution set and P is the reference point.

The larger the values of $I_C(t)$ and HV, the better the performance of the algorithm. Furthermore, based on $I_C(t)$ and HV, this paper utilizes the Wilcoxon rank-sum test and Friedman test to assess the stability of PI-CRRTCM. It should be noted that PI-CRRTCM is a initialization method specifically designed for CEWS problem. This method lacks universality, so this paper only compares it with the original MOABC population initialization method to ensure the fairness and validity of the simulation.

Fig. 6 and Table V present the statistical analysis results of the $I_C(t)$ metric and the overall algorithm performance, respectively. Fig. 6 demonstrates that under eight different scenario scales, the PI-CRRTCM method significantly outperforms the initialization strategy of the original MOABC in maintaining population diversity. The statistical results of HV and $I_C(t)$ in Table V further validate the positive effect of PI-CRRTCM on enhancing the algorithm's overall performance. The HV results of each algorithm are provided in Table A.2 of Appendix F.

TABLE V
STATISTICAL COMPARISON OF TWO METHODS IN 8 SCENARIOS

	RIni	Ini
+/-/≈		0/13/3
Ranking	1.0625	1.9375

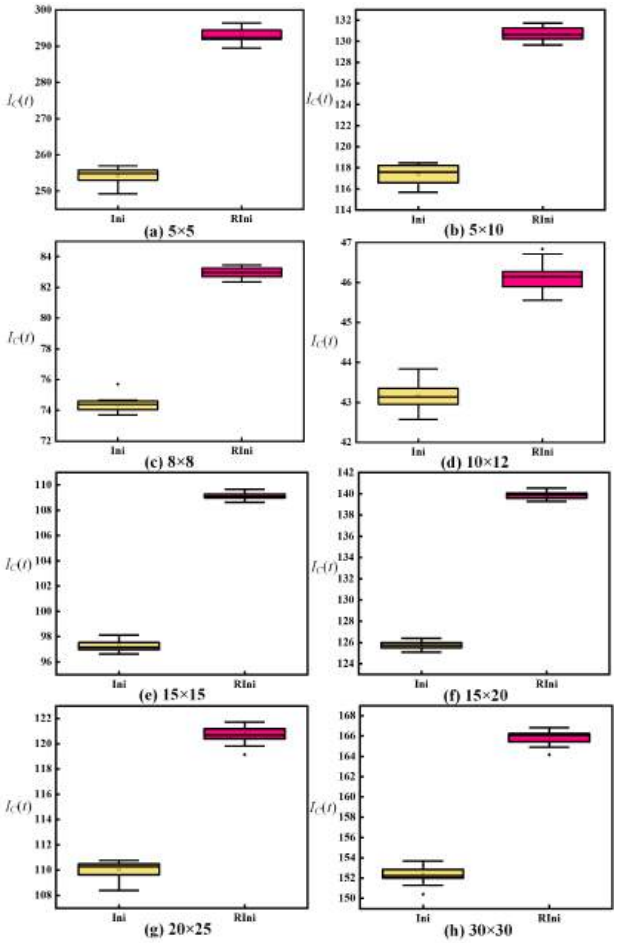


Fig. 6. Box plots comparing the $I_C(t)$ values obtained by the two initialization methods. Note that RIni denotes the proposed method, while Ini denotes the original method in the figures.

C. MLEAMO validation

MLEAMO is an individual position update strategy. To verify its effectiveness and progressiveness, algorithms of NSGA-III [46], iAM2M [47], and MOABC [36] are selected for comparison. In addition, the individual position updating strategies in the algorithms of AROA [39] (MOABC-R), Gaussian mutation beetle swarm optimization [20] (MOABC-G), and improved firefly [19] (MOABC-F) are applied to the MOABC algorithm to further compare with MLEAMO.

The Pareto fronts of different algorithms in 8 scenarios are shown in Fig. 7. The results indicate that there is no significant difference between MLEAMO and other algorithms in small-scale scenarios. As the scenario scale increases, MLEAMO exhibits better convergence performance compared to other algorithms.

To further evaluate the effectiveness of MLEAMO in improving algorithm convergence, this paper employs the HV value as the evaluation metric for Wilcoxon rank-sum test and the Friedman test. To ensure robustness, each algorithm independently runs 30 times across all test scenarios, with their HV value distributions shown in Fig. 8. The results demonstrate that MLEAMO achieves superior HV values compared to the six other algorithms across various scenarios. Table VI summarizes the statistical analysis of HV values, further confirming that MLEAMO obtains the lowest ranking values among all scenarios, indicating its optimal performance. The findings of Wilcoxon rank-sum test also corroborate this conclusion.

Meanwhile, the set coverage (SC) [48] metric $C(F1, F2)$ is also used to evaluate the effectiveness of MLEAMO in improving algorithm convergence, as calculated below

$$C(F_1, F_2) = \frac{|\{f \in F_2 | \exists f' \in F_1 : f' \prec f\}|}{|F_2|} \tag{39}$$

where F_1 and F_2 denote the Pareto solution sets of the algorithms, respectively; $C(F_1,F_2)$ =1 means that all solutions of F_2 are dominated by some solutions of F_1, while $C(F_1,F_2)$ = 0 implies that no solution of F_2 is dominated by a solution of F_1. The SC results of the algorithms are shown in Fig. 9. The results in Fig. 9 show that the minimum value of $C(7, F_i)$ for all scenarios is 0.5238, while the maximum value of $C(F_i, 7)$ is 0.3775, which indicates that MLEAMO has the best convergence for solving the CEWS model.

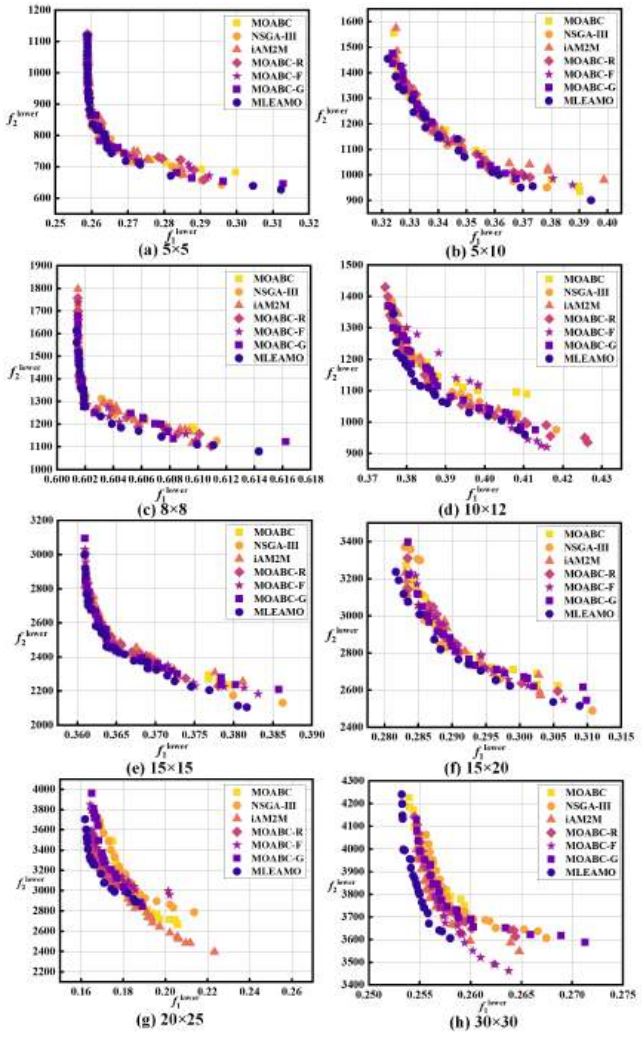


Fig. 7. Pareto front comparison of different algorithms

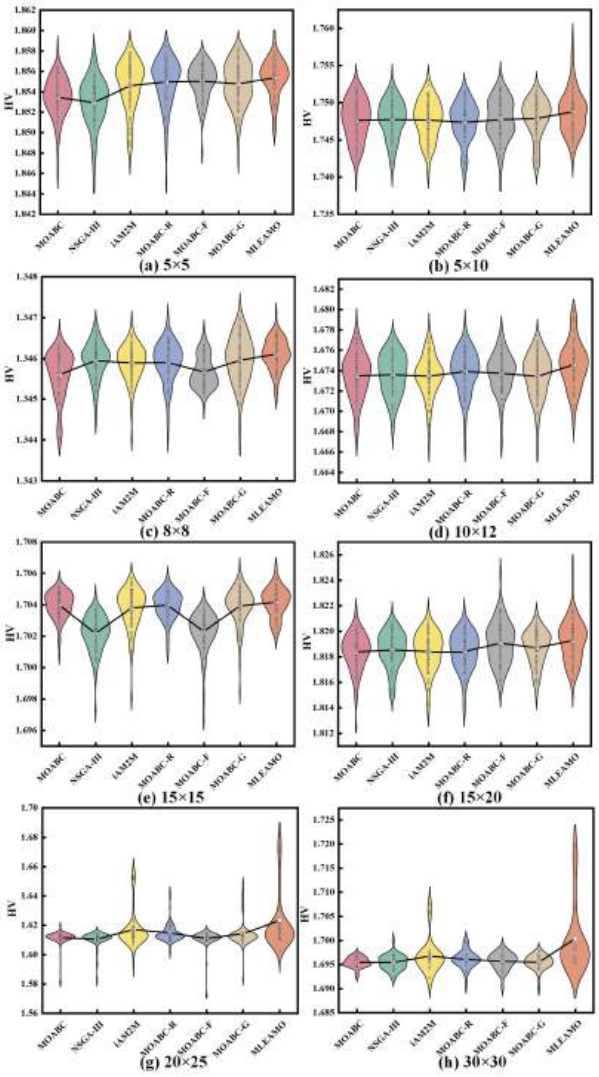


Fig. 8. HV value distribution in different scenarios

TABLE VI
STATISTICAL RESULTS OF SEVEN ALGORITHMS ACROSS DIFFERENT SCENARIO SCALES

	M1	M2	M3	M4	M5	M6	M7
+/-/≈	0/6/2	0/7/1	0/4/4	0/4/4	0/4/4	0/3/5	
Ranking	4.452	4.984	3.642	3.982	4.541	3.931	2.221

Note: Labels M1-M7 denote MOABC, NSGA-III, iAM2M, MOABC-R, MOABC-F, MOABC-G, and MLEAMO, respectively.

D. IAMOABC validation

Exploring the optimal assignment of grassroots unit resources based on a given cooperation engagement scheme is a complex task. This requires IAMOABC to have good convergence to avoid local optima in irregular solution spaces. Therefore, we demonstrate the convergence of IAMOABC based on SC and HV, and further validate its robustness through statistical analysis. Advanced multi-objective optimization algorithms such as MOBMA [49], MOEADCMT [50], and CMOEMT [51] have been selected as comparison algorithms.

Fig. 10 shows the SC results of all algorithms in different scenarios. Obviously, the SC value of IAMOABC is much higher than other compared algorithms, which fully proves that IAMOABC has superior convergence. Table VII further lists the HV values and statistical experimental results of the four algorithms. The results indicate that IAMOABC achieved the best computational performance in the vast majority of scenarios, only slightly surpassed by MOBMA and CMOEMT in one scenario. This conclusion is supported by the Friedman test ranking results, and IAMOABC obtained the smallest ranking value.

E. Comprehensive verification of BLTAA

Due to the strong coupling characteristics of the CEWS model, the existing WTA bi-level algorithm has limitations in decoupling solution. This is mainly manifested in the frequent use of repair operators to maintain the upper-level model results during the algorithm solving process, which restricts the convergence performance. To systematically verify the effectiveness and progressiveness of BLTAA, this paper com-

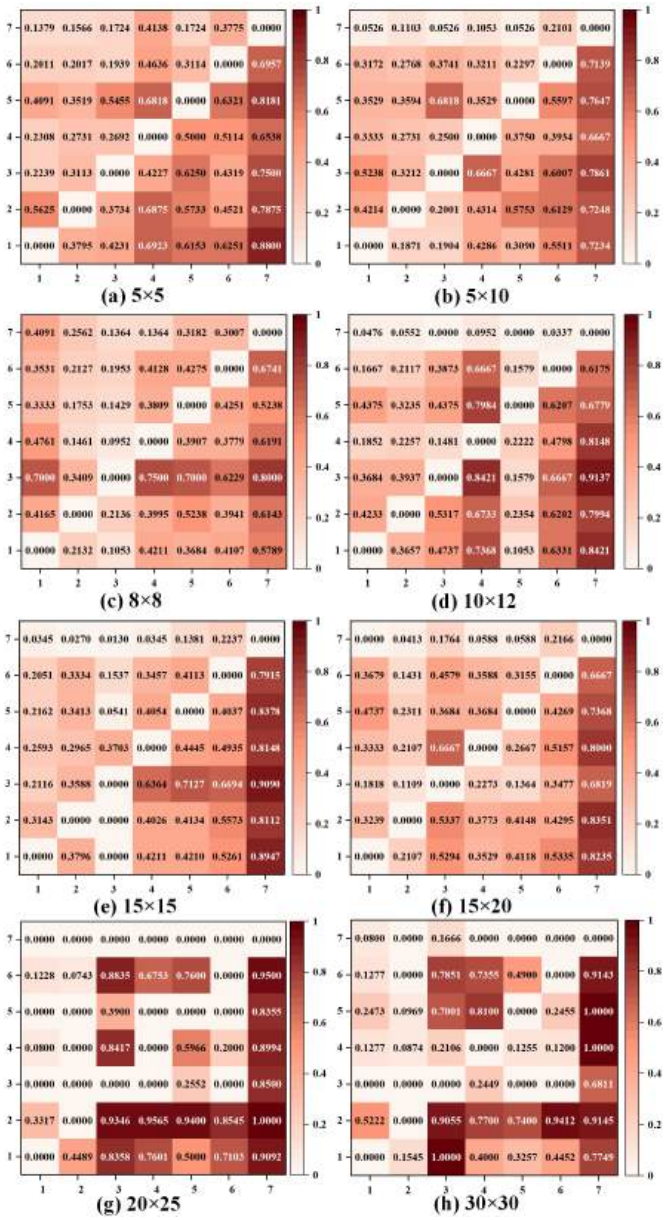


Fig. 9. SC results between algorithms in 6 scenarios. Labels 1-7 denote the Pareto solution sets obtained from MOABC, NSGA-III, iAM2M, MOABC-R, MOABC-F, MOABC-G, and MLEAMO, respectively.

bines WTA classic algorithm and advanced multi-objective optimization algorithm based on the algorithm framework in Fig. 4. The upper-level uses GA, while the lower-level uses the algorithms of NSGA-III [46], iAM2M [47], and MOBMA [49], MOEADCMT [50], and CMOEMT [51], respectively. Five comparative algorithms are constructed: GA-NSGAIII, GA-iAM2M, GA-MOBMA, GA-MOEADCMT, and GA-CMOEMT.

Fig. 11 shows the comparison for the solving results of different algorithms in 8 scenarios. The results show that the convergence of the solution set obtained by BLTAA is significantly better than that of the comparative algorithm, and this advantage becomes more prominent as the scale increases. Notably, in some scenarios, it can be observed that the resource

TABLE VII
WILCOXON RANK-SUM TEST AND FRIEDMAN STATISTICAL ANALYSIS RESULTS FOR FITNESS VALUES OF FOUR ALGORITHMS

	MOBMA	MOEADCMT	CMOEMT	IAMOABC
5×5	**1.8742**≈ (0.0027)	1.8727≈ (0.0116)	1.8701- (0.0223)	1.8734 (**0.0019**)
5×10	1.7604- (**0.0027**)	1.7413≈ (0.0048)	**1.7625**+ (0.0051)	1.7619 (0.0035)
8×8	**1.3492**+ (**0.0003**)	1.3470- (0.0022)	1.3466- (0.0017)	1.3488 (0.0005)
10×12	1.6831- (0.0072)	1.6724- (0.0049)	1.6731- (0.0125)	**1.6841** (**0.0022**)
15×15	1.7058≈ (**0.0007**)	1.7044- (0.0013)	1.7037- (0.0042)	**1.7067** (0.0009)
15×20	1.8229- (0.0066)	1.8188- (0.0272)	1.8173- (0.0513)	**1.8399** (**0.0018**)
20×25	1.6271- (0.0512)	1.6046- (0.0746)	1.6088- (0.0944)	**1.6334** (**0.0342**)
30×30	1.7191- (**0.0053**)	1.6955- (0.0227)	1.6929- (0.0573)	**1.7259** (0.0073)
+/-/≈	1/5/2	0/6/2	1/7/0	
Ranking	1.750	3.375	3.625	1.250

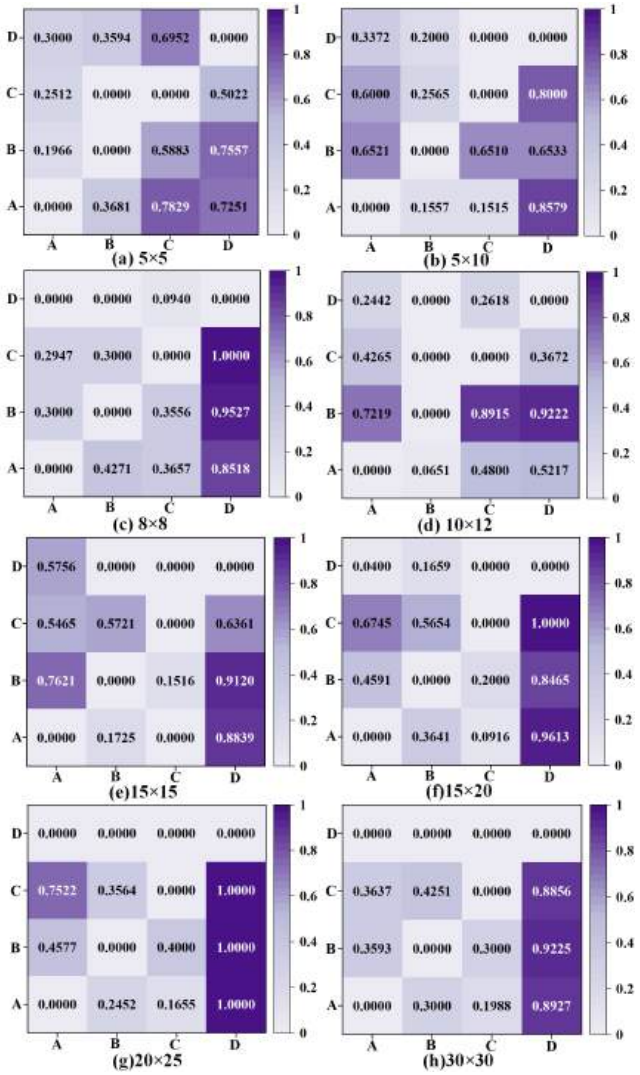


Fig. 10. SC results between algorithms in 8 scenarios. Labels A-D denote the Pareto solution sets obtained from MOBMA, MOEADCMT, CMOEMT, and IAMOABC, respectively.

cost of the solution obtained by BLTAA is higher than other algorithms. This is because the PI-CRRTCM strategy adopted by BLTAA will refine ammunition assignment based on resource reserves and target threat values. When resources are sufficient, this mechanism tends to apply stronger firepower to high threat targets, and this dynamic ammunition assignment may lead to an increase in resource consumption.

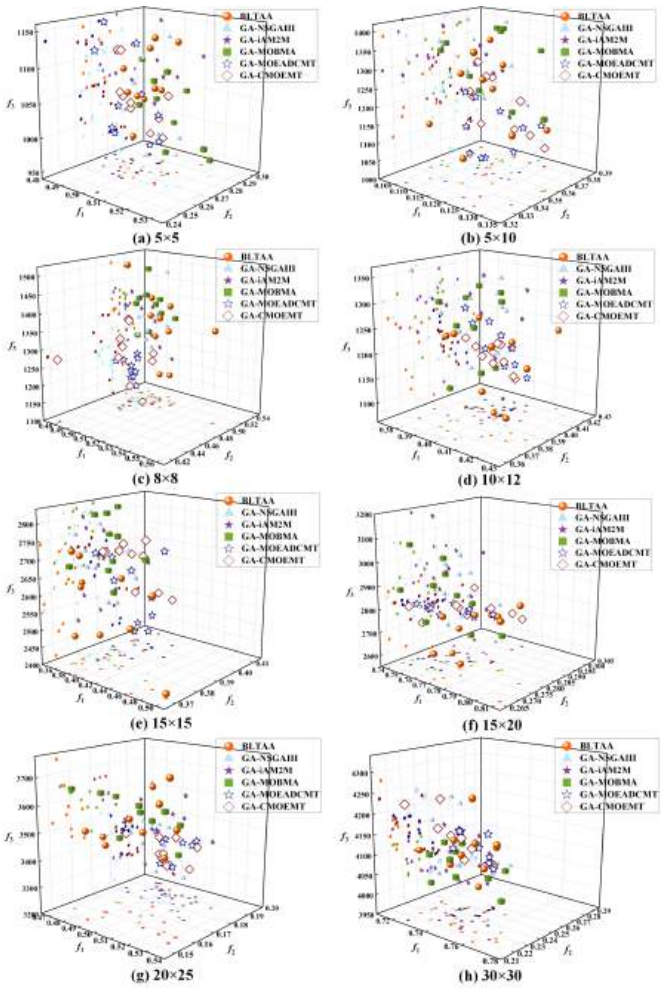


Fig. 11. Comparison of bi-level algorithms in 8 scenarios

In terms of OSC screening, the strategy screening method of BLTAA is SGSSM. Based on the solution set obtained from BLTAA, this paper compares the SGSSM method with the existing strategy screening method TOPSIS. Figs. 12(a)-(c) show that the SGSSM method outperforms the TOPSIS method in at least two objectives. Although there is a slight sacrifice in resource cost, the difference is within an acceptable range. Furthermore, Figs. 12(d)-(f) compare the OSC obtained by BLTAA and the comparison algorithm, and the results show that BLTAA has significant advantages in all three objective function values, especially in the lower-level objective functions. It is worth noting that the upper-level model objective function values obtained by SGSSM are generally equal, but this does not mean that the resulting cooperation engagement schemes are consistent. A detailed explanation is provided in Appendix G.

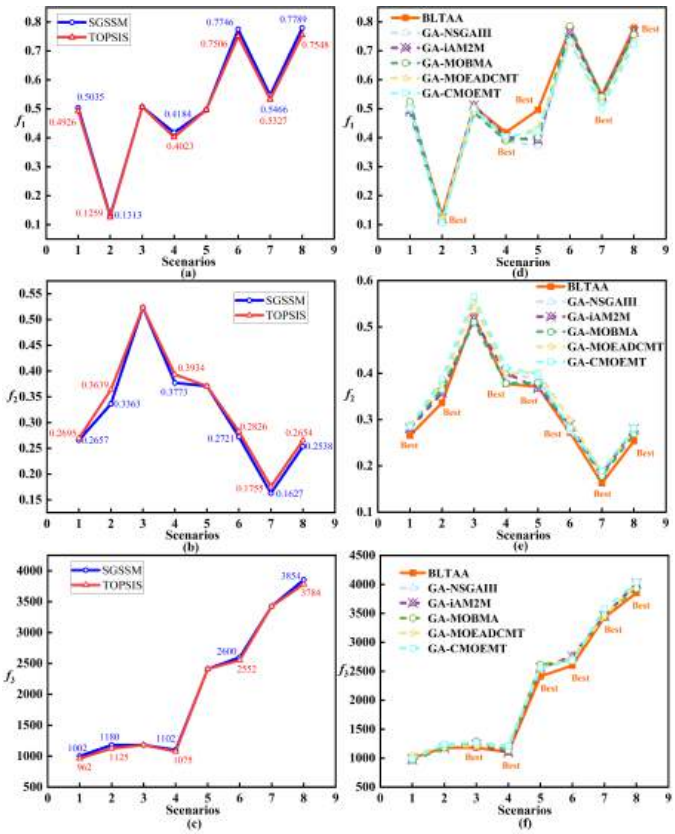


Fig. 12. Comparison of OSC results obtained by different bi-level algorithms

TABLE VIII
PERFORMANCE STATISTICS OF SIX ALGORITHMS AT DIFFERENT SCALES

	B1	B2	B3	B4	B5	B6
+/-/≈		0/20/4	0/15/9	1/11/12	1/20/3	0/19/5
Ranking	1.1250	5.1250	3.7083	2.6667	4.2500	4.3750

Note: Labels B1-B6 denote BLTAA, GA-NSGA-III, GA-iAM2M, GA-MOBMA, GA-MOEADCMT, and GA-CMOEMT, respectively.

To enhance the credibility of simulation results, this paper conducted statistical analysis on the objective function values of the CEWS model. BLTAA and the comparison algorithms are respectively executed 30 times. The results of the Wilcoxon rank-sum test and the Friedman test are shown in Table VIII. The data shows that BLTAA has significant advantages in solving the CEWS model. Especially when the scene scale expands, BLTAA can achieve higher defense benefits with less resource consumption. Tables A.3, A.4, and A.5 in Appendix F provide the objective function values obtained by each algorithm.

Computational efficiency is a key indicator for measuring algorithm performance. To evaluate the computational efficiency of BLTAA, the average computation time of each algorithm is recorded after independently running 30 times in eight different scenarios. As shown in Fig. 13, the computation time of BLTAA is at a moderate level, which is attributed to investing more computing resources into global search and solution quality optimization.

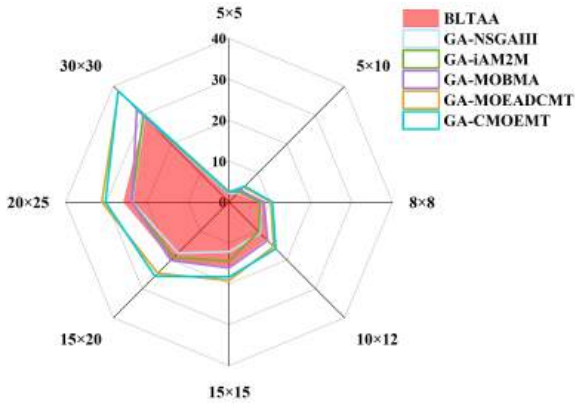


Fig. 13. Comparison of computational efficiency results obtained by different bi-level algorithms

F. Discussion

The strong coupling of the CEWS model refers to the structural dependence between the upper-level decision variable $\mathbf{X}$ and the lower-level decision variable $\mathbf{Y}$. This is a logical coupling determined by the essence of the problem, and traditional single-level optimization algorithms cannot directly handle this problem. The BLTAA proposed in this article adopts a nested optimization approach at the solution strategy level, and decouples decision variable relationships through algorithm nesting. Specifically, the master-slave iterative mechanism based on Stackelberg game transforms strongly coupled bi-level problems into serialized subproblems, achieving computability of complex problems and rationality of decision order, and ensuring that the lower-level solutions always depend on the upper-level decisions.

However, the BLTAA method heavily relies on explicit modeling or precise solution of the lower-level reaction function, which challenges the algorithm's adaptability and robustness in large-scale complex scenarios. Recently, data-driven methods such as graph neural networks have provided new ideas for modeling and optimizing complex coupled systems [52], [53], [54]. By implicitly learning the response relationships between levels, they further enhance the scalability and generalization ability of algorithms while maintaining decision order. Specifically, the paradigm of "offline training, online inference" followed by data-driven methods is key to meeting the strict real-time requirements of WTA problems.

Overall, in order to solve the WTA problems, the approach proposed in this article aims to build a basic framework that does not rely on historical data and has a transparent decision-making process, especially suitable for scenarios where data is scarce and decision-making interpretability is high. Compared to other methods, data-driven approaches are more suitable for complex battlefield environments with high real-time requirements due to their excellent time performance and powerful inference capabilities [55], [56]. It is worth noting that heuristic frameworks can provide high-quality initialization strategies for data-driven models, effectively improving their training efficiency and convergence accuracy [22], [30]. Therefore, the fusion of the two methods provides a theoretical guidance for solving real-time intelligent decision-making problems in large-scale environments.

V. Conclusions

This paper examines a new extension CEWS in military operations research. Considering the sequence of weapon system assignment and ammunition assignment, a bi-level optimization model is designed and corresponding bi-level optimization algorithm, BLTAA, is proposed in this paper. In this algorithm, an IGA is firstly introduced to solve the upper-level model. Secondly, PI-CRRTCM is proposed for maintaining population diversity. Then, MLEAMO is designed to enhance the solution exploitation ability of the algorithm. Finally, SGSSM is designed to assist the commander in making final decisions. Simulation experiments show that the proposed algorithm is effective and advanced in different scenarios.

The research is to provide a CEWS scheme without excessively considering its time performance. This is allowed in static problems, but in future research on dynamic problems, time window constraints should be added. Meanwhile, combining heuristic algorithms with deep reinforcement learning methods and validating them in open-source simulation software will be our future research direction.

Appendix

A. Information section

For the information section, with reference to Chen et al. [34] the ITC can be represented as follows:

$$I = v_{t,d}\frac{1}{D_y} + v_e\left(\frac{V_{\max}}{10} + \frac{S_{\max}}{1000}\right) \tag{A.1}$$

where D_y is the actual transmission delay of the data chain system (ms); $V_{\max}$ is the maximum transmission rate of the data chain information (Mbit/s); $S_{\max}$ is the transmission distance corresponding to the maximum transmission rate (km); $v_{t,d}$ and v_e are the indicator weights.

The IPC can be represented as follows:

$$P = \left(v_d\frac{n_d}{10} + v_b\frac{10^{-5}}{B_{\mathrm{E}}}\right) \tag{A.2}$$

where n_d is the type of message that the data chain system can handle; B_{E} is the data chain system bit error rate; v_d and v_b are the indicator weights.

The NMC can be expressed as follows:

$$N = v_g\frac{1}{t_i} + v_h\frac{n_e}{1000} \tag{A.3}$$

where t_i is the time (s) required for a new weapon system to join the datalink system; n_e is the maximum number of users that can be accommodated by the datalink system; v_g and v_h are the indicator weights, respectively.

B. Perception section

In the perception section, this paper only considers the role of radar's perception of incoming targets, therefore, based on Chen et al. study [34], the perception capability of radar can be simplified and expressed as follows:

$$S_R = \phi \cdot \frac{R^2}{4} \cdot \frac{\theta}{360} \cdot \exp\left(\frac{-0.1625 D_i}{D_0}\right) \cdot \frac{N_u^{0.05}}{\Delta_{ar} \cdot \rho_{ar}} \tag{A.4}$$

where ϕ is the detection influence coefficient in the adversarial environment; R is the maximum detection distance (km); D_i is the standard detection distance; D_0 is the maximum discovery distance; N_u is the number of targets tracked; Δ_{ar} is the localization accuracy of the target (m); and ρ_{ar} is the radar resolution.

C. Aggregation of capability indicators

To address the relationship between the capacity indicators, they are normalized based on the following satisfaction function:

$$f(a,b) = \begin{cases} \sin\left(\frac{a}{b} \cdot \frac{\pi}{2}\right), & a < b \\ 1, & a \geq b \end{cases} \tag{A.5}$$

where a is the value of the current weapon system's capability; b is the value of the capability required for the mission.

Based on a weighted summation method, we aggregate the information capability section as

$$M_i = \lambda_1 f_{\text{IT}i} + \lambda_2 f_{\text{IP}i} + \lambda_3 f_{\text{NM}i} \tag{A.6}$$

where λ_1, λ_2, and λ_3 are the indicator weights; $f_{\text{IT}i}$, $f_{\text{IP}i}$, and $f_{\text{NM}i}$ are the capability fulfillment of I, P, and N, respectively.

Based on Eq. A.5, the perception capability is normalized as

$$R_{\text{ar}}^{ij} = \begin{cases} \sin\left(\frac{S_R}{R_j^{\text{sl}}} \cdot \frac{\pi}{2}\right), & S_R < R_j^{\text{sl}} \\ 1, & S_R > R_j^{\text{sl}} \end{cases} \tag{A.7}$$

where R_{ar}^{ij} is the perception capability of weapon system i for target j; R_j^{sl} is the value of the capability required for the perception task.

D. Improved genetic algorithm: Adaptive operator

In this paper, we introduce an adaptive operator based on truncated mean, which is part of our previous research work, aiming to enhance the ability of the algorithm to jump out of the local optimum. Specifically, the population individuals are first screened based on the truncated mean and the average fitness of the population is calculated by

$$M = M_{\max}\left(\frac{t}{T_{\max}^{\text{U}}}\right)^2 \left(1 + \alpha\left(\frac{1}{2} - N(t)\right)\right) \tag{A.8}$$

where M is the number of eliminated individuals; $M_{\max}$ is the maximum number of single elimination; t is the number of current iterations; $T_{\max}^{\text{U}}$ denotes the maximum number of iterations for the IGA; α is the coefficient of influence of population aggregation; N_t is the degree of population dominance, which is defined as

$$N(t) = \frac{\sum_{i=1}^{N_p} \delta[f_i(t) < f_{\text{mean}}(t)]}{N_p} \tag{A.9}$$

$$\delta[f_i(t) < f_{\text{mean}}(t)] = \begin{cases} 1, & \text{if } \delta[f_i(t) < f_{\text{mean}}(t)] \text{ is true} \\ 0, & \text{if } \delta[f_i(t) < f_{\text{mean}}(t)] \text{ is false} \end{cases} \tag{A.10}$$

where $f_i(t)$ and $f_{\text{mean}}(t)$ are the individual fitness and the average fitness of the population at the t-th iteration, respectively; $\delta[f_i(t) < f_{\text{mean}}(t)]$ is the indicator function.

The average population fitness can then be expressed as

$$f_{\text{m}}^*(t) = \frac{\sum_{i=1}^{N_p - M} f_i(t)}{N_p - M} \tag{A.11}$$

Finally, $f_{\text{m}}^*(t)$ will serve as the standard for measuring the quality of individual populations. In the crossover stage, the adaptive operator based on $f_{\text{m}}^*(t)$ is as

$$P_c = \begin{cases} P_{c\cdot\max} - \frac{P_{c\cdot\max} - P_{c\cdot\min}}{1+\exp\left(A\cos\left(\frac{\tilde{f} - f_{\text{m}}^*}{f_{\max} - f_{\min}}\pi\right)\right)}, & \tilde{f} \geq f_{\text{m}}^*, \\ P_{c\cdot\max}, & \tilde{f} < f_{\text{m}}^*. \end{cases} \tag{A.12}$$

where $P_{c\cdot\max}$ and $P_{c\cdot\min}$ are the maximum and minimum values of the crossover probability, respectively; $\tilde{f}$ is the maximum fitness among the individuals to be crossovered; and A is the crossover adjustment factor.

In the mutation phase, the adaptive operator based on $f_{\text{m}}^*(t)$ is as

$$P_m = \begin{cases} P_{m\cdot\max} - \frac{P_{m\cdot\max} - P_{m\cdot\min}}{1+\exp\left(B\cos\left(\frac{f - f_{\text{m}}^*}{f_{\max} - f_{\min}}\pi\right)\right)}, & f \geq f_{\text{m}}^*, \\ P_{m\cdot\max}, & f < f_{\text{m}}^*. \end{cases} \tag{A.13}$$

where $P_{m\cdot\max}$ and $P_{m\cdot\min}$ are the maximum and minimum values of the mutation probability, respectively; f is the maximum fitness among the individuals to be mutated; and B is the mutation adjustment factor.

With the above formulas, the adaptive updates of crossover probability and mutation probability can be achieved. Specifically, in the maximization problem, the evolutionary probability of an individual is maximized when its fitness is less than the mean f_{m}^* and decreases exponentially when its fitness is greater than f_{m}^*. It designs different genetic operators for individuals with varying degrees of superiority and inferiority and increases the probability that inferior individuals generate new individuals while ensuring the retention of superior individuals improves the global search ability and convergence in the early and later stages of the algorithm, respectively.

E. Parameter sensitivity analysis

To ensure the performance of BLTAA, a sensitivity analysis on its key parameters is conducted. The IGA parameters in BLTAA are determined in previous work and directly adopted in this paper. For IAMOABC, MELAMO is the core strategy for enhancing its search performance, and the parameters C_p, A_P, c, μ, and ρ are the key factors that determine the algorithm's performance. This paper adopts the Taguchi method

[15] to design an orthogonal experiment with 5 factors and 4 levels to determine the optimal parameter combination for MELAMO. The levels of each factor are shown in Table A.1. The HV_{mean} value is used as the response value for orthogonal experiments. The calculation process of HV_{mean} is as follows: IAMOABC runs independently 30 times at each scenario, normalizes the obtained HV values, and takes the average value to obtain the performance indicator HV_{m}. On this basis, the average value of HV_{m} obtained from 8 different scales is applied to obtain HV_{mean}, which serves as the response value for the orthogonal experiment. The response curves and signal-to-noise ratio (S/N ratio) results of the orthogonal experiment are shown in Fig. A.1. Obviously, IAMOABC has the best computational performance when $C_p = 6$, $A_P = 12$, $c = 0.88$, $\mu = 0.34$, and $\rho = 0.12$.

TABLE A.1
DIFFERENT LEVELS OF PARAMETER VALUES

	C_p	A_P	c	μ	ρ
level 1	2	10	0.82	0.32	0.12
level 2	4	12	0.84	0.33	0.13
level 3	6	14	0.86	0.34	0.14
level 4	8	16	0.88	0.35	0.15

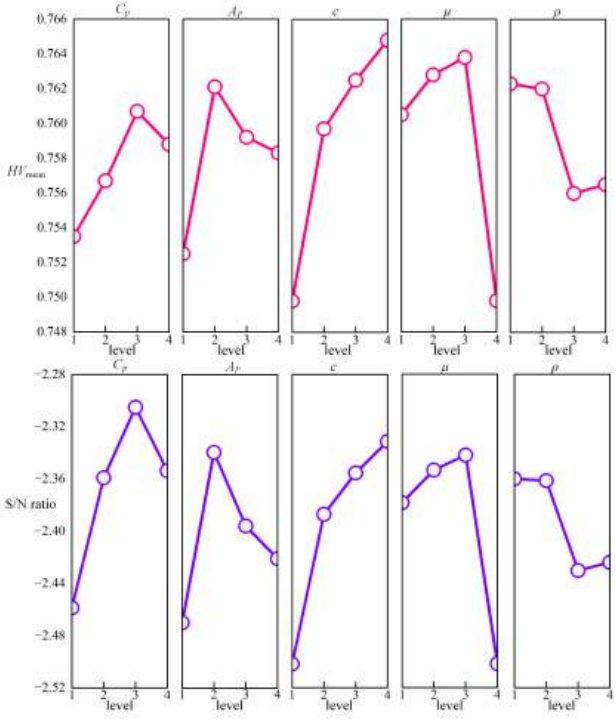


Fig. A.1. Response values and S/N ratio results of different parameter combinations

F. HV values obtained from different comparative experiments

The HV results obtained by RIni and Ini in 8 different scenarios are shown in Table A.2. The objective function values for all comparison algorithms in different scenarios are provided in Tables A.3, A.4, and A.5.

TABLE A.2
HV RESULTS OBTAINED BY RINI AND INI IN 8 DIFFERENT SCENARIOS

	RIni	Ini
5×5	**1.8531** (**0.0073**)	1.8522≈ (0.0096)
5×10	**1.7472** (0.0464)	1.7421- (**0.0025**)
8×8	**1.3461** (0.0073)	1.3454≈ (**0.0049**)
10×12	**1.6612** (**0.0611**)	1.6606- (0.0774)
15×15	1.7011 (**0.0254**)	**1.7025**≈ (0.0774)
15×20	**1.8073** (**0.0052**)	1.8035- (0.0054)
20×25	**1.6115** (0.0557)	1.6079- (**0.0271**)
30×30	**1.6889** (**0.0126**)	1.6807- (0.0351)

TABLE A.3
STATISTICS OF f_1 RESULTS OBTAINED BY DIFFERENT ALGORITHMS

	A1	A2	A3	A4	A5	A6
5×5	0.5044 (**0.0073**)	0.5036- (0.0104)	0.5038≈ (0.0108)	**0.5103+** (0.0117)	0.5029≈ (0.0103)	0.5031≈ (0.0212)
5×10	**0.1201** (0.0038)	0.1126≈ (**0.0054**)	0.1117≈ (0.0079)	0.1138≈ (0.0086)	0.1181≈ (0.0057)	0.1155≈ (0.0053)
8×8	**0.5058** (**0.0079**)	0.4929≈ (0.0174)	0.4923≈ (0.0107)	0.4937≈ (0.0093)	0.4933- (0.0105)	0.4967≈ (0.0215)
10×12	**0.4086** (**0.0054**)	0.3987≈ (0.0094)	0.3945- (0.0114)	0.4013≈ (0.0093)	0.3983- (0.0250)	0.4011- (0.0105)
15×15	**0.4084** (**0.0247**)	0.3878- (0.0353)	0.3908- (0.0506)	0.3977- (0.0284)	0.3885- (0.0725)	0.3865- (0.0231)
15×20	**0.7769** (**0.0071**)	0.7552- (0.0081)	0.7587- (0.0084)	0.7678≈ (0.0137)	0.7568- (0.0243)	0.7650≈ (0.0085)
20×25	**0.5415** (**0.0072**)	0.5217- (0.0212)	0.5227- (0.0324)	0.5215- (0.0253)	0.5225- (0.0227)	0.5177- (0.0275)
30×30	**0.7773** (**0.0055**)	0.7545- (0.0072)	0.7554- (0.0064)	0.7613≈ (0.0073)	0.7527- (0.0121)	0.7611≈ (0.0051)

Note: Labels A1-A6 denote BLTAA, GA-NSGA-III, GA-iAM2M, GA-MOBMA, GA-MOEADCMT, and GA-CMOEMT, respectively.

TABLE A.4
STATISTICS OF f_2 RESULTS OBTAINED BY DIFFERENT ALGORITHMS

	A1	A2	A3	A4	A5	A6
5×5	0.2721 (**0.0077**)	0.2728- (0.0078)	0.2730- (0.0105)	0.2799- (0.0117)	**0.2719**≈ (0.0127)	0.2737- (0.0191)
5×10	**0.3463** (0.0156)	0.3541- (**0.0155**)	0.3479≈ (0.0161)	0.3563- (0.0106)	0.3588- (0.0212)	0.3572- (0.0403)
8×8	0.5139 (**0.0075**)	0.5291- (0.0082)	0.5168≈ (0.0081)	0.5177≈ (0.0732)	**0.5029+** (0.0103)	0.5205- (0.0551)
10×12	**0.3836** (**0.0073**)	0.3858≈ (0.0175)	0.3938- (0.0114)	0.3855- (0.0117)	0.3922- (0.0314)	0.3946- (0.0225)
15×15	**0.3731** (**0.0044**)	0.3985- (0.0064)	0.3816- (0.0051)	0.3766≈ (0.0062)	0.3951- (0.0473)	0.3977- (0.0512)
15×20	**0.2757** (**0.0038**)	0.2883- (0.0089)	0.2811- (0.0048)	0.2762≈ (0.0059)	0.2855- (0.0375)	0.2821- (0.0455)
20×25	**0.1688** (**0.0244**)	0.1927- (0.0667)	0.1745≈ (0.0329)	0.1703≈ (0.0325)	0.1734- (0.0373)	0.1739- (0.0442)
30×30	**0.2597** (**0.0017**)	0.2812- (0.0155)	0.2633≈ (0.0029)	0.2629- (0.0037)	0.2713- (0.0525)	0.2802- (0.0311)

TABLE A.5
STATISTICS OF f_3 RESULTS OBTAINED BY DIFFERENT ALGORITHMS

	A1	A2	A3	A4	A5	A6
5×5	**1048** (**45.8719**)	1077- (59.9816)	1080- (48.1876)	1078- (49.8140)	1088- (49.2856)	1075- (48.5772)
5×10	**1247** (**76.4126**)	1302- (92.2090)	1305- (92.9516)	1299- (82.4554)	1325- (93.2556)	1347- (92.2719)
8×8	**1335** (**55.2042**)	1412- (65.9053)	1392- (62.8191)	1376≈ (58.4866)	1388- (61.9225)	1395- (60.7040)
10×12	**1201** (**66.0246**)	1292- (70.0471)	1259≈ (73.7130)	1248- (73.3409)	1283- (73.7876)	1288- (73.2440)
15×15	**2586** (**69.1466**)	2735- (83.3572)	2644- (92.5119)	2613- (77.7932)	2855- (89.2676)	2712- (90.2723)
15×20	**2739** (**92.8237**)	2952- (94.6427)	2825≈ (95.0877)	2786≈ (94.3738)	2847- (93.2216)	2856- (90.3583)
20×25	**3427** (**92.2935**)	3627- (87.3572)	3585- (90.2123)	3473≈ (88.8242)	3577- (95.3720)	3529- (91.5725)
30×30	**3854** (**96.3555**)	3979- (97.1722)	3922- (96.0877)	3876- (97.4271)	3952- (93.2216)	3935- (98.4417)

G. Some explanations about SGSSM

Generally, the fitness of the two individuals selected from the upper-level of BLTAA is equal. But this does not mean that their cooperative engagement schemes are consistent. For example, in the scenario of 10×12 in this paper, the upper-level of BLTAA yields two identical individual fitness values, but their cooperative engagement schemes differ significantly, as shown in Fig. A.2. This indicates that the bi-level optimization model has higher complexity and a wider range of solution space distribution.

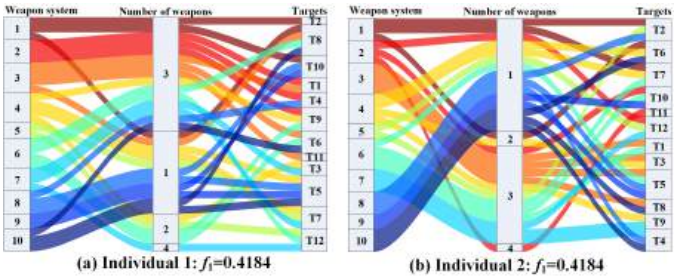


Fig. A.2. Differences between cooperative engagement schemes with equal fitness values

For the strongly coupled solution space in the bi-level model, BLTAA ensures sufficient exploration through bi-level iteration. SGSSM effectively filters the optimal solution for each iteration and ultimately obtains OSC, ensuring screening efficiency and accuracy. Compared with TOPSIS, the payoff function of SGSSM combines subjective preferences with objective weights more flexibly, better meeting the diverse needs in practical applications.

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